

Muhammad Arsalan

Data Engineer | Apache Airflow · PySpark · AWS · Azure

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github.com/Muhammad-Arsalan-Ai | muhammad-arsalan-ai.github.io/portfolio

Open to On-site/Hybrid (Karachi) & Remote (Worldwide)

SUMMARY

Data Engineer with 3+ years of experience building scalable production-grade data pipelines using Apache Airflow, PySpark, Kafka, and cloud platforms (AWS, Azure). Experienced in distributed data processing, workflow automation, real-time streaming, batch processing, and cloud-based data architecture. Proven track record delivering reliable ETL systems, optimizing pipeline performance, and maintaining high data quality across production systems.

SKILLS

Data engineering: Data Pipelines, ETL/ELT Pipelines, Apache Airflow, Data Orchestration, Workflow Automation, Apache Spark (PySpark, SparkSQL), Distributed Data Processing, Apache Kafka, Real-Time Data Streaming, Batch Processing, Data Modeling, Data Warehousing, Data Lake, Lakehouse Architecture, Delta Lake, Databricks, Snowflake, Medallion Architecture, OLAP (Online Analytical Processing), OLTP (Online Transaction Processing), Data Quality Monitoring, Data Governance

Languages & frameworks: Python, JavaScript, SQL, Flask, FastAPI, HTML, CSS, Bootstrap

Cloud & infrastructure: AWS (S3, Athena, Glue, EMR, Lambda, Kinesis, Redshift, RDS, EC2, IAM), Azure (ADF, Data Lake, Synapse Analytics), Cloud Data Architecture, Serverless Pipelines, Docker, Linux, CI/CD, Scalable Systems, Production Systems, Performance Optimization

Databases: PostgreSQL, MySQL, MongoDB (NoSQL), Qdrant (Vector DB), Redis, QuestDB

Data Analysis & Visualization: Power BI (DAX, OLAP), Streamlit, Matplotlib, Seaborn, Pandas, NumPy, Statistics

AI & LLMs: LangChain, RAG Pipelines, LLM Applications, Prompt Engineering, OpenAI API, Gemini, Hugging Face, Computer Vision

Tools & workflow: Git, Scrapy, Selenium, BeautifulSoup, Web Scraping, Jira, ClickUp, Excel, Google Sheets

PROFESSIONAL EXPERIENCE

Data Engineer, OfficeField

May 2025 – Present

- Engineered Apache Airflow-orchestrated pipelines automating real-time stock and crypto data extraction from 5 financial APIs (FMP, Nasdaq, CoinMarketCap, CoinGecko, etc), reducing data refresh cycles from daily to sub-hourly.
- Automated real-time ingestion and multi-destination storage across MongoDB, PostgreSQL, AWS RDS, and Qdrant Vector DB — processing thousands of financial records daily with zero manual intervention.
- Implemented automated data quality monitoring and validation checks across 4 production storage systems, achieving 99%+ data accuracy for downstream analytics consumers.
- Optimized workflow orchestration and Airflow task scheduling through parallelized ETL execution, improving pipeline performance and SLA compliance across real-time financial data pipelines.

Data Extraction Specialist, UpWork & Fiverr

Jul 2024 – Apr 2025

- Delivered 30+ web scraping projects using Scrapy, Selenium, and BeautifulSoup — extracting and processing millions of records across paginated, dynamic, and CAPTCHA-protected websites.
- Automated end-to-end data collection workflows, reducing client manual effort by ~80% and delivering structured datasets within agreed SLAs on every project.

- Engineered robust web scraping solutions for JavaScript-rendered and dynamically paginated websites, maintaining 98%+ extraction completeness across high-volume datasets.

Data Engineer, Xloop Digital Services (Pvt) Ltd

Feb 2023 – Jun 2024

- Architected scalable distributed data processing pipelines using PySpark and AWS Glue to transform high-volume customer datasets powering real-time recommendation systems.
- Deployed a real-time Kafka streaming pipeline on AWS EC2 handling highthroughput event streams, integrated with AWS Glue for automated schema cataloging — reducing data availability lag from hours to under 5 minutes.
- Built a Python-based multi-database ETL pipeline to extract, clean, and segment customer data by region — loading USA records into MySQL and UK/India records into PostgreSQL, with structured JSON metadata for downstream processing.
- Designed DAX-optimized Power BI dashboards consolidating multi-source financial complaint datasets, enabling self-service analytics for 5+ stakeholders and eliminating recurring manual reporting.

PROJECTS

AWS Real-Time Data Streaming, Stack: AWS EC2, Kafka, AWS Glue, Athena, S3

github.com/Muhammad-Arsalan-Ai/AWS-Real-Time-Data-Streaming

- Built an end-to-end streaming pipeline producing CSV data via Kafka on EC2, consuming it into S3 for centralized storage.
- Implemented AWS Glue Crawler for automated schema cataloging, enabling immediate Athena querying without manual schema management.

ETL Pipeline: Excel to MySQL & PostgreSQL, Stack: Python, MySQL, PostgreSQL, JSON

github.com/Muhammad-Arsalan-Ai/ETL-Pipeline--Excel-to-MySQL-and-PostgreSQL

- Engineered a complete ETL pipeline processing customer data from Excel, with automated cleaning and country-based segmentation.
- Loaded USA records into MySQL and UK/India records into PostgreSQL, exporting structured JSON metadata for downstream processing.

Counselor Recommendation System, Stack: PySpark, Redis, Docker

https://github.com/Muhammad-Arsalan-Ai/Counselor_Recommendation_System

- Built a recommendation system matching counselors to clients by specialization, using PySpark for large-scale data processing.
- Deployed Dockerized ETL and matching services with cronjob-based scheduling, enabling automated, low-latency in-memory lookups via Redis.

EDUCATION

Bachelor Of Computer Science

2019 – 2023

- Sir Syed University of Engineering and Technology

CERTIFICATES

- **Data Engineering Bootcamp — Emeritus:** 2023
- **Generative AI with Large Language Models — Coursera / DeepLearning.AI:** 2024
- **Certified Artificial Intelligence Developer — PIAIC:** 2023
- **Introduction to Data Analytics — LinkedIn Learning:** 2024